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Professional Summary

Senior Software Engineer with over 15 years of experience delivering enterprise software, cloud-native solutions and business-critical platforms across the energy, retail, telecommunications, sports media and financial services sectors.

Over the past several years, I have specialized in AWS cloud engineering, event-driven architectures and modern software delivery practices. I have extensive experience designing, developing and supporting secure, scalable and highly available systems using C#, .NET, TypeScript, React, PostgreSQL, SQL Server, AWS and Azure DevOps. Currently working within the Australian energy sector, I collaborate closely with traders, business stakeholders, architects and cross-functional engineering teams to deliver technology solutions that support critical business operations and customer outcomes.

My experience spans cloud-native application development, reporting platforms, enterprise integrations, payment systems, ETL and data warehouse solutions, production support and operational excellence. I enjoy solving complex technical problems, improving engineering practices and building solutions that create measurable business value. I regularly leverage AI-assisted development tools to improve productivity, accelerate delivery and enhance software quality.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

AWS Certified Developer – Associate

AWS Certified AI Practitioner

Skills

- AWS (Fargate, ECR, ECS, VPC management, S3, Step Functions, Appsync, IAM, CDK)
- C#.Net, .Net Core, VB.Net, Java, Python, ASP.net (MVC), TypeScript, React.
- MS Sql Server 2005/2008/2014/2017/2019, MySql, Oracle, DynamoDb, Postgres Sql.
- Extensive Web Service and Window Service development in .net framework.
- MSMQ (Microsoft Messaging Queue), RabbitMq, SQS, Kafka.
- Restful Web Api design and development, GraphQL, API Gateway.
- Crystal Reports, MS Excel (for reporting).
- OR mapping (Hibernate 3.0), Entity Framework, TypeORM, Prisma.
- MS Visual Source Safe, TortoiseSVN, Github, GitFlow, DataDog monitoring.
- Working knowledge of SSIS and SSRS, Athena.

- Avoka platform (online forms) and Kofax, KTM (OCR solutions).
- TDD mainly used NUnit and Jest.

Professional Experience – brief

- Shell – 4 years – Cloud focus, Agile delivery, AI centric
- Catapult – 1 year – Event driven solutions
- Tennis Australia – 4 years – API development. Microservices.
- Decipha (Aus Post) – 1 year – OCR solutions, image processing.
- Premier Tech (NAB) – 4 years - Online payment platform.
- Uelco (Telstra) – 6 months – ETL processes.
- Genie Technologies – 4 years – Retail data processing – ETL.
- LoneStar Tech – 2 years – CRM software – desktop/web development.

Professional Experience – detail

Shell (<https://www.shellenergy.com>)

08/2022 – working

Senior AWS Engineer

Responsibilities:

- Designed and delivered **cloud-native** applications supporting energy trading and B2B business operations using C#, .NET, TypeScript, React and AWS.
- Architected and implemented event-driven processing solutions using Amazon S3, EventBridge, **ECS Fargate and Docker**.
- Built customer reporting platforms using **Amazon Athena**, **AWS Step Functions** and Parquet datasets stored in S3.
- Implemented Infrastructure as Code using AWS CDK to support automated and **repeatable cloud deployments**.
- Led **CI/CD improvements** within a large mono repository, ensuring only impacted applications triggered deployments and reducing unnecessary pipeline execution.
- Enhanced release governance through Azure DevOps approval workflows and **separation-of-duties** controls for production deployments.
- Improved cloud security through migration from IAM users to IAM roles and implementation of **IAM-based PostgreSQL authentication**.
- Implemented Lambda-based authorization for **AWS AppSync** APIs.
- Introduced cancellation capabilities for long-running customer reports, improving customer experience and operational efficiency.
- Enhanced observability through **Datadog monitoring** and alerting improvements.
- Participated in **AWS Well-Architected review** sessions with AWS architects covering security, reliability, performance and operational excellence.
- Contributed to disaster recovery planning and operational resilience initiatives.
- Drove cloud cost optimization initiatives through Lambda optimization, environment rationalization and **infrastructure right-sizing**.
- Developed and presented an AI-powered document processing proof of concept using **AWS Bedrock**, converting PDF documents into structured business data.
- Regularly utilize **AI-assisted development tools** to improve delivery speed, solution quality and engineering productivity.

Achievements:

- Improved security, no more IAM users with long lived credentials. No more storage of passwords in the secret manager, instead now IAM roles are used to connect to DB.

- Cost saved by decommissioning staging CRM, staging resources now points to test db. Right sizing of resources, i.e. no need of massive db clusters for small datasets. Moving files from Standard S3 to IA buckets.
- Moved on premises servers to EC2. Lift and shift mostly.
- Encouraged the management to use **Athena instead of loading parquet files** to Db and then read it from there.
- Brown bag sessions for AI ingestion in daily development lifecycles. **AI powered OCR solution** is awaiting productionized for cross section teams.

Tennis Australia ([HYPERLINK "http://www.tennis.com.au/"](http://www.tennis.com.au/) [HYPERLINK "http://www.tennis.com.au/"](http://www.tennis.com.au/) [HYPERLINK "http://www.tennis.com.au/"](http://www.tennis.com.au/) [HYPERLINK "http://www.tennis.com.au/"](http://www.tennis.com.au/))
01/2018 – 12/2021

Senior Software Developer

Responsibilities:

- Designed, implemented and maintained a system that is responsible for showing content on TV during **live matches** of AO (Australian Open). Covered 4 AO (2018/2019/2020/2021).
- **Improved processes** to receive data from SMT(sports media tech) and HE(Hawkeye).
- Implemented restful **apis** to receive match data from SMT and HE during live matches.
- Implemented apis to expose calculated stats to third parties (TV/Web/Media). Channel 9/Channel 7/ EuroSport used these apis to display content on their **broadcast**.
- Moved the whole **infrastructure to AWS**, implemented tasks in clusters. Fargate tasks.
- Moved TARS (Tennis Australia Ratings System) in TA servers from external vendors.
- Designed and implemented a **data sharing mechanism** between Tennis Australia and UTR (Universal Tennis Rating).
- Collaboration with data scientists, graphic designers, external developers/solution architects and media personals.
- Maintained tennislab.com.au website for National Tennis Centre.

Achievements:

- Led a team of developers to achieve success during live events, despite being short on resources.
- Chased HE for SLAs failure, TA got back a substantial amount of money.
- Improved design, code, deployment procedures, logging and repo structure.
- We as a team won an award from media industry for the most innovative stat producers for tennis.

Decipha ([HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/) [HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/) [HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/) [HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/) [HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/) [HYPERLINK "http://www.decipha.com.au/"](http://www.decipha.com.au/))
04/2016 – 12/2017

Senior Analyst Developer

Responsibilities:

- Worked on a solution (online forms) of Australia Post for transferring lands.
- Implemented modules to display critical information on **dashboards** (dashing.io).
- Sql Upgrades from sql2005 to sql2014.
- MCC (Melbourne Cricket Club) solution for memberships, a paper based solution using **kofax**.
- Upgraded **Avoka** TM(Transaction manager).
- Provided technical help to **peer and junior developers**. Cost estimation for different projects and CRs, Worked with IT support staff to fix prod issues.
- Implemented unit tests in critical modules to encourage test driven development.

Achievements:

- Implemented imaging solution for MCC.

- Upgraded Avoka platform.
- Improved SVN structure for groovy scripts.
- Added new dashboards in production to monitor db sizes.
- Supported an online solution that got **Digital disruption National Award for 2016**. (Disclaimer: I have not designed or built it from scratch, but I have maintained it for one year)

Premier Technologies ([HYPERLINK "http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/) [HYPERLINK "http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/) [HYPERLINK "http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/) [HYPERLINK "https://linkly.com.au/"](https://linkly.com.au/) [HYPERLINK "http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/) [HYPERLINK "http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/))
 09/2012 – 04/2016

Senior Software Developer

Responsibilities:

- Integrated American Express Checkout (**Amex's eWallet**) into payment page (**ASP.net MVC**) for hosted and non-hosted merchants. Designed and developed the processes for merchant onboarding for new and existing merchants.
- Integrated MasterPass (**MasterCard's eWallet**) into payment page for hosted merchants.
- Upgraded the whole platform to work with **UTC** dates instead of local dates. Code (**Web Services, Window Services, and Class Libraries**) and database changed moreover migrated/converted historic records to new dates.
- Designed and developed a dedicated **Windows Service** (Monitor Service) to automate rebooting process of critical production servers during off peak hours. Following functionalities were also added once the service was matured.
 - Ability to restart crashed window services, in case of failure sends out alerts to infrastructure team to troubleshoot.
 - Monitoring system stats (disk space in log folders and memory consumed by processes).
 - Automate deployment process (deploying dll and config files to server during reboot).
- Developed eForm solution (backend only), which allows users to design **web forms** as per their needs and publish them to consumers to submit data. Added reporting ability for merchants to download the submitted forms from back office.
- Upgraded **payment gateway (cardGate)** from version 4 to 5, transferred the related software and hardware from Windows 2003 to Windows 2008. Implemented **disaster recovery** process.
- Maintained and improved the existing shift placement system for casual staff, which is currently serving **CBA (Commonwealth Bank of Australia)** employees in different parts of Australia.
- Redesigned and implemented software components to read and write messages from **MSMQ (Microsoft Messaging Queues)** in a secure way.
- Developed the payment page (for iPhone and Android users) for Australia post's bill payment App (**APDM – Australia Post Digital Mail**) (**ASP.net MVC**).
- Maintained **IVR (Interactive Voice Response)** code base, checked in legacy code in SVN.
- Improved the production deployment process by using **shell scripts**, introduced the concept of before and after cutover tasks. Added critical steps in CC (Change Control) process.
- **Functional and Technical** specs, **UML diagrams** (sequence and flow) and support documents for **ECommerce** team were added in documentation process.

Achievements:

- Reduced the time and effort of manual reboot process, automated the whole exercise. Automated production deployment tasks.
- Improved the deployment processes (scripting instead of manual tasks) to production servers to decrease the downtime for customers.
- Saved costs by maintaining old systems (original plans were to rewrite module(s) from scratch.)

Utelco Systems Pte Ltd (<http://www.utelcosystems.com.au/> [HYPERLINK](#)

["http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/) [HYPERLINK](#) ["http://www.utelcosystems.com.au/"](http://www.utelcosystems.com.au/)

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01/2012 – 07/2012

ETL Developer - Vertica/ SQL Server (Contract)

Responsibilities:

- Documented, designed and implemented the **ETL and ELT** routines to populate the existing star schema for Mobile Data Analytics in **Telstra's UAT** environment.
- Worked on **LDM (Logical Data Modeling)** for current projects, identification of potential entity types and relation between them.
- Analyzed key business areas, tracking true source systems to enhance the scope of network reports, to figure out the **integration points** between different service providers.
- Identified the **gaps** in the business processes and highlight them to client for their review in weekly review meetings.
- **Source to target** mapping, using an innovative technique to map the target back to the source so that missing information from a source system to warehouse architecture can be identified as quickly as possible.

Genie Technologies ([HYPERLINK "http://www.idmsoftware.com/"](http://www.idmsoftware.com/) [HYPERLINK](#)

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05/2007 – 09/2011

Senior Software Engineer (Team Lead)

Responsibilities (DEV):

- Followed technical specs and developed spreadsheet module using **Win-Forms** of the Loss Manager application.
- Added ability to extract data from application in different formats including **text (.csv), html, image and excel**.
- Designed and implemented **drill down/ drill-through** functionality to link results of one module with another. It is useful when a user wants to drill into details of a particular analysis.
- Operated in **Agile** Methodologies with a focus on **Scrum**.

Responsibilities (ETL):

- I was part of the group, which was selected for **ETL** training, passed the course with distinction and made a member of ETL team (I was only developer in the start, when I left team was grown to 6 members).
- Analyzed and researched different tools to meet Genie's ETL requirements. In the end designed and developed custom ETL program (**C# windows and web services**).
- For 3 years **hired and led** a team of five developers and successfully extracted, transformed and loaded data for UK clients into data warehouse, using the same tool which I developed initially. Added more functionality into the ETL program.
- Assigned tasks, monitor timelines and **mentor junior developers and code reviews**.
- Analyzed client's **raw** data based on the documents provided and defined rules to extract transactions from multiple sources. This involved Client **communication** before and after deployment.
- Developed mechanisms for **memory management** of huge data to address memory issues.
- Developed a **scheduler** to automate transformation on daily, weekly, or monthly basis. It picks data from the configured source directory, processes it, stores the transformed data into output directory and notifies on process completion.

Responsibilities (Data Warehouse):

- Analyzed the reporting requirements for different scenarios and created **summary tables** to increase query performance for huge (500 Million) data for a single store. This needs to be done while prudently recognizing **measures** and **hierarchies**.
- Applied indexing on columns after carefully analyzing their data. Like for columns with more **distinct** values got **bitmap** indexes.
- Implementation of **business layer** to group the logical fields together and allow user to see the data as per their business definition.
- Employed **archiving** of data technique to remove the un-necessary **partitions** from the star on the end of each month.
- Documented and applied **Backup** and **Disaster Recovery** plans.
- Facilitated clients to design complex **ad-hoc reports** according to business needs.

Achievements:

- Hired and trained the transformation (ETL) team, developed processes, documents and standards for new team to follow.
- Served 8 clients (5 EU, 3 US) with ETL jobs, this was huge achievement as initially Genie has decided to outsource the ETL process.
- It was not only the communication skills but also superior knowledge of business domain, which enables me to act as a bridge between onshore and offshore teams.

LoneStar Technologies, Lahore

06/2005 – 08/2007

Software Engineer

LONESTAR provides a wide range of IT services including business Intelligence, dashboards, IT Solutions and Services, RFID solutions, eCommerce development and BackOffice Solutions.

Responsibilities:

- Developed several modules of the desktop application using **VB.Net** which includes development of **UI** and handling of saving/retrieving from database.
- Designed and developed the data layer of web version using **Hibernate3.0 (O/R mapping)**.
- Developed Reports using Seagate **Crystal Reports 9**.
- Converted the SQL server database schema to **mySql**.

Achievements:

- With the help of O/R mapping of Hibernate I made business development team free from concerns of databases. This means that they only need to call DAO functions, created by me, for saving and retrieving data, rest was my job to communicate with database.
- I got 10 out of 15 monthly awards, which were given on performance basis.

Education

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|--------------------------|--------------|---------------------------------------|
| • MSc (Computer Science) | 2001 to 2004 | Quaid-e-Azam University (QAU). |
| • BSc (Math, Physics) | 1999 to 2001 | Punjab University (PU). |

Activities and Interests

- Exploring ways to enhance productivity with ChatGPT for personal tasks.
- Preparing for Azure Certifications.
- DIY back garden improvements.

References

- References will be provided on request.